

Making STEAM: Build Your Custom Electric Guitar

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Figure 1: Undergraduate students from the music department's 2023 First Year Seminar class with completed guitars

ABSTRACT

By teaching the Science, Technology, Engineering, Art, and Math embedded in the objects they create, STEAM-based digital fabrication projects equip the next generation of leaders and problem solvers with the tools to make change and build the future. The Make Your Electric Guitar project has been tested successfully with undergraduate (Figure 1), middle, and high school students (Figure 2) since 2019. As students design and develop their guitar, they not only build a tool for self-expression but also learn iterative design, fabrication, and the physics of sound and circuitry and leave with a working electric guitar.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Performing arts; Education; Collaborative learning; Physical sciences and engineering; Engineering; Computer-aided design; Physical sciences and engineering; Electronics.

KEYWORDS

STEAM Education, Design, Fabrication, Making, Materials, CAD, Circuits

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1 INTRODUCTION

STEAM Studio was founded in 2016 by a collective of UNC Asheville faculty and staff from the engineering and sculpture departments who saw value in collaboration, found inspiration in each other and the limitless potential of maker-based education to elevate the student experience and create positive change in our communities. Since its inception, in addition to supporting students at UNC Asheville, STEAM Studio has hosted outreach activities for Western North Carolina K12 students.

Developed in 2019, the guitar curriculum was first used in a summer camp format for STEAM Studio's community outreach program, SkillSet. Six middle school and high school girls spent one week immersed in the design, development, fabrication, and testing of their guitars. The project has been taught as a summer camp for five years and has been adapted to be a First Year Seminar experience for undergraduate students at UNCA with collaborating faculty from the music department. As the project continues, we find new ways of refining the curriculum and are ready to share it more broadly.

2 COURSE DESCRIPTION

Students design and build an electric guitar at UNC Asheville's STEAM Studio using a mix of digital fabrication and traditional hand tools. Students ideate and prototype designs in cardboard, then convert the 2-dimensional cardboard designs into 3-D computer models using Fusion 360. In digital space, the guitar models are refined and tool-paths are created for the digital fabrication machines. After dimensioning rough-sawn lumber in the wood shop, students cut the guitar body out using the CNC router. Custom pickguards cut using the waterjet. During the assembly process, students solder the electrical circuits and fasten the neck, body, saddle, and pickguard to make a fully functional, custom electric guitar. The final assembly is ripe with challenges and opportunities for developing problem-solving skills.

3 COURSE OBJECTIVES

By the end of the course, students should be able to:

- recognize and implement the principles of design and basic ergonomics
- name and identify the function of the components of an electric guitar, including circuit elements
- explain the physics of sound and electromagnets
- sketch and develop two-dimensional designs



Figure 2: Students from the first SkillSet summer camp in 2019

Table 1: Metadata in tabular format.

Summary	Students design and build a custom electric guitar and learn basic 3D modeling, digital fabrication methods, and basic electric circuits
Classification(s)	Digital Fabrication
Audience	Undergraduate, Middle, and High School
Dependencies	Students should be comfortable with computers and have the ability to perform tasks with their hands, e.g., assembly with a screw driver
Prerequisites	No pre-requisites
Strengths	Students learn 3D modeling and other STEM concepts while being creative and are highly motivated by the project outcome
Weaknesses	Generally, the 3D modeling has a steep learning curve, the project requires specialized tools that are found in many academic makerspaces, but are otherwise not accessible
Variants	Students can explore a variety of guitar body shapes through 3D modeling, any 3D modeling software could be used, a laser cutter could be used in lieu of a waterjet
Assessment	Project completion

- convert 2D design to 3D model
- understand how to create a tool path for the CNC router
- safely use power tools and hand tools to cut, shape, and finish wood and metal
- assemble components of an electric guitar

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